Fine Motor Skills of Clinical Laboratory Scientists

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Introduction

- Hand functioning is a fundamental fine motor skill in the Clinical Laboratory Science (CLS) profession.
- Fine motor skills include in-hand manipulation (IHM), reach and grasp patterns.
- Fine motor skills are related to performance of work-related tasks (Pont, Wallen, & Bundy, 2009) such as the Purdue Pegboard Test of the 1940s.
- Lack of studies that identify the fine motor skills that CLS professionals need.

Research Question

What are the essential fine motor skills of Clinical Laboratory Science professionals?

Methods

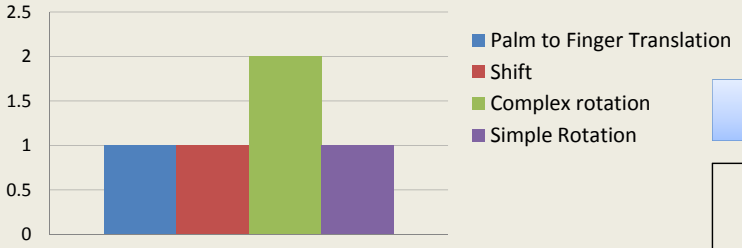
- The data was collected and recorded on VideoANT software (http://ant.umn.edu/), through observations of one CLS practitioner performing routine reagent and sample preparations.
- "Units of doing " identified as the activity that CLS professional is doing and the purpose it serves.
- Classification of fine motor skills within each activity.
- Use of both hands together were also classified as symmetrical or asymmetrical, measuring percent of time spent using one hand only, two hands symmetrically or two hands asymmetrically.

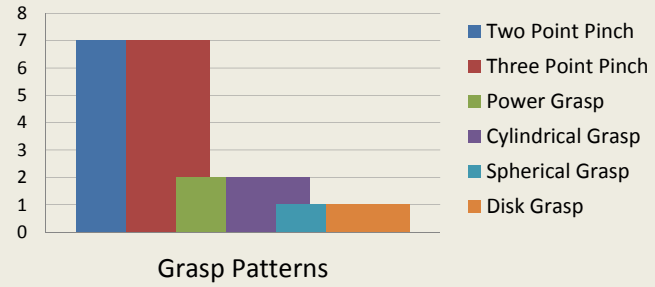


Results

- Common grasps used within the "units of doing" of CLS practice include two point pinch (pincer grasps) and the three point pinch.
- These two hand functions are used throughout the preparatory work of one clinical laboratory scientist more often than others.
- Within each observational sample or "unit of doing", bimanual asymmetry was more common than symmetrical hand movements.

Frequency of Observed CLS Fine Motor Skills

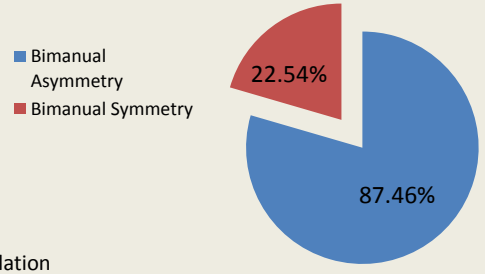




Discussion

- The common fine motor skill used in various "units of doing" are two point pinch (pincer grasps) and the three point pinch. It was commonly used for placement or holds of test tubes, slides or pipetting.
- Within each observational sample or "unit of doing", bimanual asymmetry was more common than symmetrical hand movements.
- In many of the practices, the clinical laboratory scientist manipulated multiple equipment in both hands to complete the task that also required dexterity, precision and strength.

Bimanual Asymmetry vs. Bimanual Symmetry



Implications

- These findings help describe the motor demands within recurring units of clinical science practice.
- Further implications may include adaption of tools or equipment by medical companies for increased productivity or adapting the needs of individual's disability.
- These results will also serve to provide a structure of criteria in admission of the clinical laboratory science field.
- This study mainly focused on describing what fine motor skills are needed, however other possible directions of the study may include assessing balance, eye-hand coordination and strength of the hands.

Limitations

A limitation to this study is that only one clinical laboratory scientist was observed. To better assess the essential fine motor skills, more clinical laboratory scientists and observers should be included in the study to assess confounding variables.

References

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